

DATA VORTEX 2026 · ROUND 1 PHASE 2

Insight report

Team SE7EN · what the three queries actually say about the Social Engine
Tables: 12,000 posts, 1,500 users, 1 May 2024 – 30 April 2025

The short version

Follower count is not how this network works. Per-post engagement is essentially the same for a 12,000-follower account and a 37,000-follower account. Platforms are almost tied on volume; Instagram is slightly ahead on shares, which is the only reason it wins average total engagement. The “anomalous” small accounts in the top 10% of total engagement are mostly people who posted a lot, plus a handful who really do punch above their audience. If the rebuilt Social Engine ranks creators by followers, it will rank the wrong thing.

Question	What we asked SQL	What came back
E3	Highest average total engagement by platform	Instagram 4,040.02 (Facebook has more posts)
M2	High vs low followers, engagement per post	4,000.91 vs 4,003.01 (gap 0.05%)
H4	Followers < 5,000 and top 10% of total engagement	15 accounts; #1 of them is 5th in the whole user base

Finding 1 — Instagram wins on shares, not on volume

Facebook is the volume leader by one post (2,074 vs YouTube 2,073). That is not a platform ranking, it is a coin flip. Average total engagement is a real, if small, split:

platform	posts	with likes	avg likes	avg shares	avg comments	avg total
Instagram	1,989	1,693	2,500.93	1,040.84	499.80	4,040.02
YouTube	2,073	1,747	2,517.77	1,011.83	504.38	4,031.84
Facebook	2,074	1,755	2,528.86	984.17	506.94	4,015.99
Reddit	2,031	1,742	2,488.11	1,002.23	511.18	4,003.94
Twitter	2,049	1,725	2,437.69	1,005.39	506.13	3,951.91

E3 output. Instagram is first because of shares (1,040.84), not likes. Facebook actually has the highest average likes (2,528.86) and the lowest average shares (984.17).

The Instagram–Twitter gap is 88 points, about 2.2% of the overall mean (4,001.99). Nobody here is a different sport. What is useful is the mix: if the engine is trying to spread a post, Instagram is the network; if it is trying to collect reactions, Facebook and YouTube are slightly kinder. Reddit leads comments (511.18) and still sits fourth on the total, so comment volume is not the lever.

Finding 2 — A large audience does not buy engagement

This is the result we did not expect to be this clean. The high-follower group is 742 accounts, mean 37,402 followers. The low-follower group is 758 accounts, mean 12,756 followers. Almost a 3× gap on the thing we split on. On engagement per post they are the same number.

group	users	posts	with eng.	avg followers	avg eng/post	vs overall
High (>= 25,000)	742	5,925	4,982	37,402	4,000.91	-1.07
Low (< 25,000)	758	6,075	5,204	12,756	4,003.01	+1.03

M2 output, trimmed. High-follower posts sit 1.07 below the table mean; low-follower posts sit 1.03 above it. The 2.10-point gap is 0.05% of 4,002.

Likes, shares and comments are each flat across the split as well (likes 2,496 vs 2,488, shares 1,001 vs 1,013, comments 504 vs 504). Nothing is hiding in the mix. The Pearson correlation between follower_count and per-post engagement, on the 10,186 posts with a recorded like, is 0.003. That is a zero.

Two readings, and we cannot tell them apart from this table alone. Either the Social Engine does not distribute posts by audience size, or the follower column is not a live audience — it is a label that was generated independently of the engagement columns. Phase 1 already found likes independent of platform, month and timestamp format. M2 extends that: likes are independent of the author's follower count too. For a ranking or recommendation rebuild, treat follower_count as a profile attribute, not as a predictor.

Finding 3 — The small accounts in the top 10% are mixed, not a ring

131 of 1,500 users have fewer than 5,000 followers. If total engagement were independent of audience size — which M2 says it is, at post grain — about 13 of those 131 should land in the top decile by chance. We got 15. That is not an outbreak. It is the same distribution with a long tail of posting volume.

user_id	location	followers	posts	avg eng.	total eng.	pctile
user_fgjkkrie	Lyon, France	2,211	14	4,477.86	62,690	99.73
user_uerv85na	Rome, Italy	1,824	16	4,527.77	58,861	99.07
user_hdas0iau	Rio de Janeiro, Brazil	1,620	16	3,937.00	55,118	98.33
user_r7eg1rac	Houston, USA	2,052	12	4,546.00	54,552	98.13
user_6wra58f7	Johannesburg, South Africa	4,953	14	4,122.08	53,587	97.87
user_kbdvf8d6	Tokyo, Japan	4,645	10	4,849.00	48,490	96.00
user_67hyf45u	Vancouver, Canada	898	11	4,762.70	47,627	95.20
user_5oe5t3js	London, UK	3,151	13	4,316.45	47,481	95.13
user_cqaitbuv	Johannesburg, South Africa	3,382	11	4,740.30	47,403	95.00
user_rr1uzkql	Milan, Italy	1,069	14	3,909.33	46,912	94.53
user_n0ok02rt	Dubai, UAE	2,531	18	3,324.00	46,536	94.26
user_0aacyvtz	Rome, Italy	1,452	11	4,466.70	44,667	93.06
user_hvqqyzoi	Melbourne, Australia	3,213	12	3,983.64	43,820	92.06
user_06v0exkg	Barcelona, Spain	552	12	3,969.91	43,669	91.79
user_txrypced	Rome, Italy	2,479	10	4,289.60	42,896	91.06

H4 output. Overall post mean is 4,001.99. Rows with avg eng. below that are in the top 10% because they posted often, not because each post is hot.

The list splits. user_kbdvf8d6 (Tokyo, 4,645 followers, 10 posts, average 4,849) and user_67hyf45u (Vancouver, 898 followers, average 4,763) are rate stories: few posts, each one heavy. user_n0ok02rt (Dubai, 2,531 followers, 18 posts, average 3,324) is a volume story: the personal mean is below the table mean, and the account still clears the top decile because 18 posts is the far tail of activity (median user has 8). user_hdas0iau and user_rr1uzkql sit in the same bucket.

The headline row is user_fgjkkrie in Lyon. 2,211 followers, 14 posts, every post measurable, total 62,690. That is the fifth-highest total in the entire 1,500-user table. The four accounts above it are user_zqv2vrf5 (Vancouver, 13,531 followers, 22 posts, 81,326), user_nfo3ih5u (Barcelona, 40,429, 19 posts),

user_0irp4abu (Toronto, 44,206, 17 posts) and user_2ytzut7i (Mumbai, 17,064, 14 posts). Two of the top five are not even in the “high follower” band from M2. The leaderboard of total engagement is a leaderboard of who posted, not of who has the audience.

user_06v0exkg is the smallest audience on the H4 list: 552 followers, Barcelona, 12 posts, 43,669 total. That is not proof of fraud. Under M2 it is what the generator does. Flag it for a human if the engine has a trust queue; do not auto-punish it for “too much engagement per follower”. That ratio is meaningless here.

What we would put back into the Social Engine

- Stop using follower_count as a quality or ranking signal. It does not move likes, shares or comments.
- If a network has to be picked for distribution, pick Instagram for share-heavy posts. The edge is small. Do not pick Facebook just because it has 2,074 rows.
- A “high impact, low follower” alert should look at average engagement and post count separately. Totals alone will just surface the most active small accounts.
- Keep NULL likes as NULL in the analytical store. Filling them would create the follower-engagement relationship this pack shows does not exist.

Limits

Engagement is a sum of three counts the source file happened to contain. There is no view, click, or dwell-time column, so we cannot talk about reach. The date window is one year and the five platforms are almost balanced (χ^2 against uniform, $p = 0.66$ in Phase 1), which is convenient for E3 and a little too tidy for a live network. H4's 15 rows are a descriptive tail, not a classification model. We did not score “suspicious”; we listed who meets the brief.

All three statements are in 1_SQL_Queries.pdf and sql/01_queries.sql. The numbers in this report are the result grids from those statements, not figures typed in by hand.